

# Sharp Saturation of the Annular Coefficient-Envelope Rate

Bin Wang

July 2026

## Abstract

The mass-limited rate of RH-305 is optimal at the level of its coefficient information. We truncate the actual deterministic anchor sequence at order  $N$  and set every later model coefficient to zero. The resulting sequence satisfies the same cap envelope with parameter  $M_N = 48q^2(q_*/q)^N$ , while its annular mismatch is exactly the target tail. Both its  $H^\infty$  and  $H^2$  norms are of order  $M_N^{-\kappa(\rho)}/\log M_N$ . Thus neither the power nor logarithmic factor can be improved from the coefficient envelope alone. The truncated sequence is not asserted to be a power-sum sequence of any noisy spectral complement.

## 1 Truncated target family

Fix  $0 < \rho < \rho_*$  and put  $r = q_*/q > 1$ ,  $y = q_*\rho < 1$ . For  $N \geq 2$  define

$$\tau_n^{(N)} = \begin{cases} a_n, & 2 \leq n \leq N, \\ 0, & n > N, \end{cases} \quad M_N = 48q^2r^N.$$

The unified target envelope gives, for every  $n \geq 2$ ,

$$|\tau_n^{(N)}| \leq M_N q^{n-2}.$$

The corresponding logarithmic mismatch is

$$g_N(z) = - \sum_{n>N} \frac{a_n}{n} z^n.$$

## 2 Information-class sharpness

**Theorem 2.1.** *For each fixed  $0 < \rho < \rho_*$  there are constants  $0 < c_\rho < C_\rho < \infty$  such that, for  $X = H^\infty(\rho)$  and  $H^2(\rho)$ ,*

$$c_\rho \frac{y^N}{N} \leq \|g_N\|_X \leq C_\rho \frac{y^N}{N}.$$

*Equivalently,*

$$\|g_N\|_X = \Theta_\rho \left( \frac{M_N^{-\kappa(\rho)}}{\log M_N} \right), \quad \kappa(\rho) = \frac{\log(1/y)}{\log r}.$$

*Proof.* The upper bounds follow from  $|a_n| < 48q_*^n$  and the geometric estimates

$$\sum_{n>N} \frac{y^n}{n} = O_\rho(y^N/N), \quad \left( \sum_{n>N} \frac{y^{2n}}{n^2} \right)^{1/2} = O_\rho(y^N/N).$$

For the lower bound, let  $n_N$  be the first odd integer greater than  $N$ ; then  $n_N \leq N + 2$  and

$$\frac{|a_{n_N}| \rho^{n_N}}{n_N} = \frac{y^{n_N}}{n_N(1 + \lambda^{-n_N})} \geq c_\rho \frac{y^N}{N}.$$

This one coefficient bounds the  $H^2$  norm. For  $H^\infty$ , take the odd projection  $(g_N(z) - g_N(-z))/2$ , whose norm is at most  $\|g_N\|_\infty$ ; at positive radius all its odd target-tail terms have the same sign, so the same first term gives the lower bound. Finally  $M_N = 48q^2r^N$  converts  $y^N/N$  to the displayed mass scale.  $\square$

### 3 Protocol and firewall

The computation records rigorous lower and geometric upper bounds for the infinite constant-48 model tails at  $\rho = 1.41$  and  $N = 20, 40, 80$ ; it does not truncate an infinite sum silently. The numerical model checks the rate only.

*Remark 3.1.* The family  $\tau^{(N)}$  is a coefficient array satisfying the same envelope. No multiset in  $|\mu| \leq q$ , no noisy operator, and no physical spectral power-sum realization is constructed. Therefore the theorem proves sharpness only for the coefficient-envelope information class, not for the actual noisy complement. Gates A–E remain false/open.